

6.4. The Extended Golay Code

- Used by the Voyager and Galileo space probes
 - Golay code: the perfect $(23, 12, 7)$ -code
 - Extended Golay code: a $(24, 12, 8)$ -code
- } 3-error correcting

The construction is based on the properties of a matrix B .

Consider the field $(\mathbb{Z}_{11}, +, \cdot)$.

Exercise: Determine the quadratic residues (the squares) in $\mathbb{Z}_{11}$, that is, for which $n \in \mathbb{Z}_{11}$ is $n = a^2$ for some $a \in \mathbb{Z}_{11}$.

$$B' = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \end{matrix} \\ \begin{matrix} 11 \times 11 \end{matrix} & \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}$$

$$B = \begin{pmatrix} B' & \begin{matrix} 1 \\ \vdots \\ 0 \end{matrix} \\ \begin{matrix} 1 & 1 & \dots & 1 & 0 \end{matrix} & \end{pmatrix}_{12 \times 12}$$

$$\mathbf{B} = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \end{pmatrix}$$

6.3.2 Lemma The matrix $\mathbf{B}$ has the following properties.

- (α) $\mathbf{B}^T = \mathbf{B}$
- (β) $\mathbf{B}^2 = \mathbf{I}_{12}$ (over $\mathbb{F}_2$)
- (γ) Every row of $\mathbf{B}$ has weight 7, except the bottom row which has weight 11.
- (δ) The distance between any 2 distinct rows of $\mathbf{B}$ is 6.
- (ϵ) Each row and column of $\mathbf{B}$ has odd weight.

α, γ, ϵ by observation.
Proof of δ and β :

For $1 \leq i < i' \leq 11$, if we look at the positions of the 1s in rows i and i' , then we see that there are exactly 4 positions in common.

Thus, for $1 \leq i < i' \leq 11$, rows i and i' of $\mathbf{B}$ are at distance 6 (because the weight of each of these rows is 7) and are orthogonal (because 4 is even). For $1 \leq i \leq 11$, row i and row 12 are also at distance 6, so δ holds, and row i and row 12 of $\mathbf{B}$ are orthogonal (because row i has exactly six 1s in columns 1 to 11). Finally, since each row of $\mathbf{B}$ has odd weight, the dot product of any row with itself is 1. Thus, since $\mathbf{B}^T = \mathbf{B}$, we have $\mathbf{B}^2 = \mathbf{I}_{12}$. So β holds. $\square$

The extended Golay Code, denoted C_{24} , is the linear code with generating matrix $G = (I_{12} \ B)$.

6.3.3 Theorem [Properties of C_{24}] The code C_{24} has the following properties:

- (a) C_{24} has length 24 and dimension 12.
- (b) C_{24} has parity check matrix $H = G^T = \begin{pmatrix} I_{12} \\ B \end{pmatrix}$.
- (c) Another parity check matrix for C_{24} is $H' = \begin{pmatrix} B \\ I_{12} \end{pmatrix}$.
- (d) Another generating matrix for C_{24} is $G' = \begin{pmatrix} B & I_{12} \end{pmatrix}$.
- (e) Any two rows (distinct or not) of G are orthogonal.
- (f) C_{24} is self-dual: $(C_{24})^\perp = C_{24}$.
- (g) The distance of C_{24} is 8, so C_{24} is a 3-error correcting code.
- (h) C_{24} has one word of weight 0, namely 0_{24} , one word of weight 24, namely 1_{24} , and every other word has weight 8, 12, or 16 (in fact, the weight distribution for the codewords of C_{24} is

weight	0	4	8	12	16	20	24
# words	1	0	759	2576	759	0	1

but we don't need this and won't be proving it).

Proof (a) is clear. (b) H is a 24×12 matrix with linearly independent columns, and since $B^2 = I_{12}$ (see Lemma 6.3.2 (β)), we have

$$GH = \begin{pmatrix} I_{12} & B \end{pmatrix} \begin{pmatrix} I_{12} \\ B \end{pmatrix} = I_{12}^2 + B^2 = I_{12} + I_{12} = 0.$$

(c) H' is a 24×12 matrix with linearly independent columns, and

$$GH' = \begin{pmatrix} I_{12} & B \end{pmatrix} \begin{pmatrix} B \\ I_{12} \end{pmatrix} = B + B = 0.$$

(d) G' has the correct size, its rows are linearly independent, and

$$G'H = \begin{pmatrix} B & I_{12} \end{pmatrix} \begin{pmatrix} I_{12} \\ B \end{pmatrix} = B + B = 0.$$

(e) G is a generating matrix and $G^T = H'$ is a parity check matrix, so $GG^T = GH' = 0$ (as in (c)). So every row of G is orthogonal to every column of G^T .

(f) Since $G_{C^\perp} = H_G^T$, a generating matrix for $(C_{24})^\perp$ is $(H)^T = (I_{12}^T \ B^T) = (I_{12} \ B) = G$ (see Lemma 6.3.2 (α)), so C_{24} and $(C_{24})^\perp$ have equal generating matrices.

(g) There are two steps:

(1) The distance of C_{24} is 6, 7 or 8.

Proof By Lemma 6.3.2 (γ), there are rows of $\mathbf{G}$ with weight 8, so $d(C_{24}) \leq 8$.

Suppose there exists a codeword c with $0 < \|c\| \leq 5$. Using our two generating matrices,

$$c = u_1 (\mathbf{I}_{12} \ \mathbf{B}) = u_2 (\mathbf{B} \ \mathbf{I}_{12})$$

for some u_1 and $u_2 \in \mathbb{F}_2^{12}$. Write $c = (c_1 \mid c_2)$ where c_1 and c_2 have length 12. Since $\|c\| \leq 5$, at least one of c_1, c_2 has weight ≤ 2 . But $c_1 = u_1$ and $c_2 = u_2$. So one of u_1 or u_2 has weight ≤ 2 . Thus c is the sum of either 1 or 2 rows of a generating matrix. However, by Lemma 6.3.2 (γ) and (δ), any such c must satisfy $\|c\| \geq 6$, contradiction.

(2) If $w \in C_{24}$, then $\|w\|$ is a multiple of 4.

Proof Let r_i denote row i of $\mathbf{G}$. Any codeword w can be expressed as the sum of at most 12 rows of $\mathbf{G}$, so $w = r_{i_1} + \cdots + r_{i_m}$, where $m \leq 12$. Perform induction on m :

If $m = 0$ the result is trivial. If $m = 1$ then $\|w\| = \|r_{i_1}\| = 8$ or 12 (see Lemma 6.3.2 (γ)).

If $m \geq 1$, assume the hypothesis is true for m and consider $m + 1$.

We know $w = (r_{i_1} + \cdots + r_{i_m}) + r_{i_{m+1}} = c + r$, where $r = r_{i_{m+1}}$ and c is a codeword which is a sum of m rows of $\mathbf{G}$. By inductive hypothesis $\|c\|$ is a multiple of 4, and so is $\|r\|$ (see Lemma 6.3.2 (γ)). Also since the rows of $\mathbf{G}$ are mutually orthogonal (by (e) above), $c \perp r$ (see Lemma 4.6.4). Thus, by Lemma 2.2.5 $\|w\| = \|c + r\| \equiv \|c\| + \|r\| \equiv 0 + 0 \pmod{4}$. So, by induction, (2) is true.

It follows from (1) and (2) that $d(C_{24}) = 8$.

(h) By Lemma 6.3.2 (ϵ), the sum of all rows of $\mathbf{G}$ is $\mathbf{1}$, so $\mathbf{1} \in C_{24}$. Thus, if $c \in C_{24}$, then $c + \mathbf{1} \in C_{24}$. Hence the number of words of weight r equals the number of words of weight $24 - r$. Clearly, there is 1 word of weight 0 and 1 word of weight 24. There are no words of weight 4 (as $\delta = 8$), so there are no words of weight 20. Thus, all words have weight 0, 8, 12, 16 or 24 (recall that we have already seen that the weight of any word in C_{24} is a multiple of 4). $\square$

6.4. Decoding the Extended Golay Code

Recall that the extended Golay code C_{24} , is the linear code with generating matrix $G = (I_{12} \ B)$.

$$B = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \end{pmatrix}$$

Notation:

w	Received word
c	Closest codeword to w
e	Error pattern ($e = c + w$)
$e = (e_1 \mid e_2)$	e_1, e_2 of length 12
θ_j	The word of length 12 with a 1 in the j th position and zeros elsewhere
b_j	The j th row of matrix B
$G = (I_{12} \ B)$	The generating matrix for C_{24}
$H = \begin{pmatrix} I_{12} \\ B \end{pmatrix}$ and $H' = \begin{pmatrix} B \\ I_{12} \end{pmatrix}$	Parity check matrices for C_{24}
s	The syndrome wH
s'	The syndrome wH'

6.4.1 Lemma The syndrome $s = e_1 + e_2B$ and the syndrome $s' = sB = e_1B + e_2$.

Proof We have

$$s = wH = eH = e_1I_{12} + e_2B \quad \therefore s = e_1 + e_2B.$$

Similarly,

$$s' = wH' = eH' = e_1B + e_2I_{12} \quad \therefore s' = e_1B + e_2$$

and so

$$\begin{aligned} s' &= e_1B + e_2I_{12} \\ &= e_1B + e_2B^2 \\ &= (e_1 + e_2B)B \\ &= sB. \end{aligned}$$

6.4.2 Lemma If u is a non-zero word of length 12, then $\|uB\| \geq 8 - \|u\|$.

Proof Since C_{24} has distance 8, we have $\|uG\| \geq 8$. But $G = (I_{12} \ B)$, so

$$8 \leq \|uG\| = \|u(I_{12} \ B)\| = \|(u \mid uB)\| = \|u\| + \|uB\|.$$

Thus, $\|uB\| \geq 8 - \|u\|$.

Consider 4 cases for error patterns of weight ≤ 3 .

$e = (e_1, e_2)$ where each of e_1, e_2 is of length 12

① e_1 weight ≤ 3 with $e_2 = \underline{0}$

$$\text{Consider } eH = e \begin{pmatrix} I_{12} \\ B \end{pmatrix}$$

② e_1 weight ≤ 2 with $\|e_2\| = 1$

$$\text{Consider } eH = e \begin{pmatrix} I_{12} \\ B \end{pmatrix}$$

③ $e_1 = \underline{0}$ with e_2 weight ≤ 3

$$\text{Consider } eH' = e \begin{pmatrix} B \\ I_{12} \end{pmatrix}$$

④ $\|e_1\| = 1$ with e_2 weight ≤ 2

$$\text{Consider } eH' = e \begin{pmatrix} B \\ I_{12} \end{pmatrix}$$

The following 4 lemmas consider the possible distributions of errors in error patterns of weight at most 3.

6.4.3 Lemma If $\|e\| \leq 3$, then the following are equivalent.

- (a) $e_2 = 0$.
- (b) $e = (s \mid 0)$.
- (c) $\|s\| \leq 3$.

6.4.4 Lemma Let $j \in \{1, 2, \dots, 12\}$. If $\|e\| \leq 3$, then the following are equivalent.

- (a) $e_2 = \theta_j$.
- (b) $e = (s + b_j \mid \theta_j)$.
- (c) $\|s + b_j\| \leq 2$.

6.4.5 Lemma If $\|e\| \leq 3$, then the following are equivalent.

- (a) $e_1 = 0$.
- (b) $e = (0 \mid s')$.
- (c) $\|s'\| \leq 3$.

6.4.6 Lemma Let $j \in \{1, 2, \dots, 12\}$. If $\|e\| \leq 3$, then following are equivalent.

- (a) $e_1 = \theta_j$.
- (b) $e = (\theta_j \mid s' + b_j)$.
- (c) $\|s' + b_j\| \leq 2$.

6.4.7 Algorithm IMLD for the Extended Golay Code, C_{24}

- (a) Compute the syndrome $s = wH = w \begin{pmatrix} I_{12} \\ B \end{pmatrix}$.
- (b) If $\|s\| \leq 3$ then $e = (s \mid 0)$. Stop.
- (c) If $\|s + b_j\| \leq 2$ for some row b_j of B then $e = (s + b_j \mid \theta_j)$. Stop.
- (d) Compute the second syndrome $s' = sB$.
- (e) If $\|s'\| \leq 3$ then $e = (0 \mid s')$. Stop.
- (f) If $\|s' + b_j\| \leq 2$ for some row b_j of B then $e = (\theta_j \mid s' + b_j)$. Stop.
- (g) If e is not yet determined, then more than 3 errors have occurred so request retransmission.

IMLD for the binary Extended Golay Code C_{24}

The Golay Code C_{23}

& The Extended Golay Code C_{24}

$$\mathbf{B} = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \end{pmatrix}$$

- Let θ_i be the length 12 weight one word with a 1 in position i .
- Receive w . Compute the syndrome $s = w\mathbf{H} = w \begin{pmatrix} \mathbf{I}_{12} \\ \mathbf{B} \end{pmatrix}$.
- If $\|s\| \leq 3$ then $e = (s \mid 0)$.
- If $\|s + b_i\| \leq 2$ for some row b_i of $\mathbf{B}$ then $e = (s + b_i \mid \theta_i)$.
- Compute the second syndrome $s\mathbf{B}$.
- If $\|s\mathbf{B}\| \leq 3$ then $e = (0 \mid s\mathbf{B})$.
- If $\|s\mathbf{B} + b_i\| \leq 2$ for some row b_i of $\mathbf{B}$ then $e = (\theta_i \mid s\mathbf{B} + b_i)$.
- If e is not yet determined, more than 3 errors have occurred so request retransmission.

Exercise: Suppose codewords of C_{24} are sent over a BSC and $w = 000\ 000\ 100\ 000\ 001\ 100\ 010\ 111$ is received. The syndrome $s = w\mathbf{H} = 100\ 110\ 101\ 011$. Determine the most likely codeword sent.

IMLD for the binary Extended Golay Code

The Golay Code C_{23}

& The Extended Golay Code C_{24}

C_{24}

$$\mathbf{B} = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \end{pmatrix}$$

- Let θ_i be the length 12 weight one word with a 1 in position i .
- Receive w . Compute the syndrome $s = w\mathbf{H} = w \begin{pmatrix} \mathbf{I}_{12} \\ \mathbf{B} \end{pmatrix}$.
- If $\|s\| \leq 3$ then $e = (s \mid 0)$.
- If $\|s + b_i\| \leq 2$ for some row b_i of $\mathbf{B}$ then $e = (s + b_i \mid \theta_i)$.
- Compute the second syndrome $s\mathbf{B}$.
- If $\|s\mathbf{B}\| \leq 3$ then $e = (0 \mid s\mathbf{B})$.
- If $\|s\mathbf{B} + b_i\| \leq 2$ for some row b_i of $\mathbf{B}$ then $e = (\theta_i \mid s\mathbf{B} + b_i)$.
- If e is not yet determined, more than 3 errors have occurred so request retransmission.

Exercise Determine the syndrome $s' = s\mathbf{B}$ that corresponds to the error pattern $e = 000 \ 000 \ 000 \ 001 \ 000 \ 000 \ 000 \ 011$.

6.5. The Golay Code

6.5.1 Definition The Golay Code C_{23} is obtained by deleting the last bit from each codeword of C_{24} .

We saw that every word in C_{24} has even weight. It follows that C_{24} is the extended code of C_{23} .

6.5.2 Theorem C_{23} has the following properties:

- (a) C_{23} is a linear code. A generating matrix is $(I_{12} \ \hat{B})$, where $\hat{B}$ is obtained from the matrix B , as defined above, by deleting the last column.
- (b) C_{23} is a $(23, 12, 7)$ -code.
- (c) C_{23} is a perfect 3-error correcting code.

6.5.3 Algorithm IMLD for the Golay Code C_{23}

- (a) Form $w^* = (w \mid 0)$ or $(w \mid 1)$ such that $\|w^*\|$ is odd.
- (b) Find the closest codeword $c^* \in C_{24}$ to w^* using Algorithm 6.4.7.
- (c) Remove the last bit from c^* to obtain $c \in C_{23}$.

This algorithm clearly works unless $d(c^*, w^*) > 3$.
We have assumed $d(c, w) \leq 3$, so therefore $d(c^*, w^*) \leq 4$.
Since c^* has even weight and w^* has odd weight, $d(c^*, w^*)$ is odd. Hence $d(c^*, w^*) \leq 3$.